

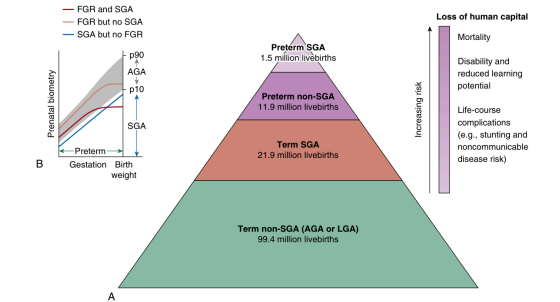


Fig. 20.1 (A) Global Burden of Small and Vulnerable Newborns in 2020. (B) Potential in utero growth trajectories reflecting the presence or absence of growth restriction and timing of birth, which impact weight per se is no longer used as a risk threshold. Instead, appropriateness of weight for gestational age provides more granular identification of programming risk in infants born preterm or at term. AGA, Appropriate for gestational age; FGR, fetal intrauterine growth restriction; LGA, large for gestational age. (Adapted from Testa et al., *Long-term renal disease after prematurity or fetal growth restriction: who is at risk?* *Nephrol Dial Transplant*. 2020;35:1087–1090 and Lavin JE, Ohuma EO, Bradley E, et al. Small babies, big risks: global estimates of prevalence and mortality for vulnerable newborns to accelerate change and improve counting. *Lancet*. 2023;401(10389):1707–1719.^{70,71})

Fig_20_1

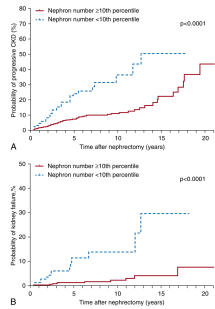


Fig. 20.4 Risk of adverse outcomes after nephrectomy among kidney tumor patients with nephron number above or below the 10th percentile for age. Low nephron number of retained kidney (<10th percentile for age and sex in healthy subset) predicted progressive chronic kidney disease (CKD) (A) and kidney failure (B) (log-rank $P < 0.0001$ for both). From Davis A, Muller AL, Alexander MP, et al. An improved method for estimating nephron number and the association of resulting nephron number estimates with chronic kidney disease outcomes. *J Am Soc Nephrol*. 2023;34(12):2758–2770.⁷²

Fig_20_4

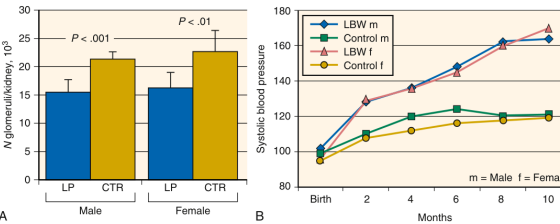


Fig. 20.7 Fetal programming of hypertension in growth-restricted rats (induced by maternal low-protein diet). (A) Total number of glomeruli per kidney at 8 weeks of age and (B) systolic blood pressure throughout life, in male ($n = 7$) and female ($n = 6$) offspring from low-protein pregnancies and in male ($n = 7$) and female ($n = 7$) control rats. CTR, Control diet; LBW, low birth weight (growth restricted); LP, low-protein diet. (Adapted from Vehaskari VM, Aviles DH, Manning J, et al. Prenatal programming of adult hypertension in the rat. *Kidney Int*. 2001;59(1):238–245.¹⁶)

Fig_20_7

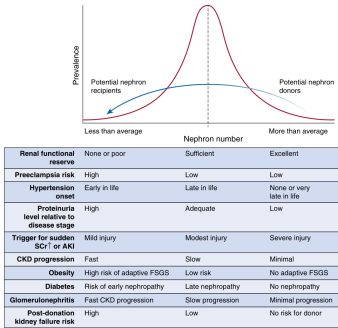


Fig. 20.2 Nephron endowment follows a normal distribution across the population. Individuals with a low nephron number have a limited capacity to tolerate superimposed kidney insults and to compensate for nephron loss and are therefore at risk of developing progressive kidney disease. People with higher nephron numbers are likely more resistant to developing a critical loss of kidney function in the setting of kidney injury because they have an abundant nephron reserve. AKI, Acute kidney injury; CKD, chronic kidney disease; FSGS, focal segmental glomerulosclerosis; SCD, sudden cardiac death. From Layke VA, Pyle AD, Tisher HV, et al. Nephron endowment as a therapeutic target to maximize kidney lifespan. *Nat Rev Nephrol*. 2022;18(6):771–783.⁷³

Fig_20_2

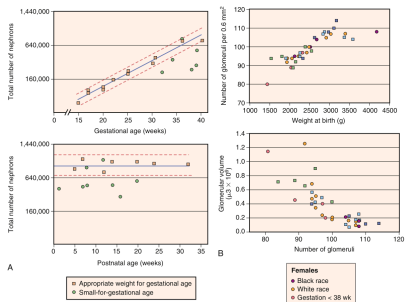


Fig. 20.5 Effect of intrauterine growth restriction on nephron number in humans. (A) Nephron number at birth in relation to gestational age (upper panel) and lack of postnatal catch-up in nephron number (lower panel). (B) Relationship between birth weight and glomerular number (upper panel) and between glomerular number and glomerular volume (lower panel). From Sampedro AL, Lopez LF, Sampedro AL, et al. The effect of intrauterine growth retardation on the development of renal nephrons. *Br J Obstet Gynaecol*. 1992;99(290–301)⁷⁴ and B from Morbach R, Heyges L, Henkel M, et al. *Kidney Int*. 2020;98(7):770–777.⁷⁵

Fig_20_5



Fig. 20.8 Architecture of wild type (WT), fibroblast growth factor 7 (FGF7) null mutant, vitamin A-deficient, and protein-deficient mouse kidneys. Embryonic day 15.5 ureteric tree tracings for each condition (to scale). N is the maximum branching generation number, and Glom represents the mean glomerular count (minimum of three kidneys studied per condition). In each case, glomerular count is reduced compared with the wild type. The number of branching generations is specific to each condition. (From Sampogna RV, Schneider L, Al-Awqati Q. Developmental programming of branching morphogenesis in the kidney. *J Am Soc Nephrol*. 2015;26(10):2414–2422.⁷⁶)

Fig_20_8

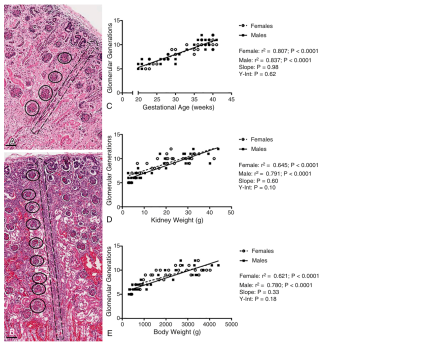


Fig. 20.3 Representative images of medullary rays (dotted line) and generations of glomeruli (circled) in hematoxylin-eosin-stained sections of renal cortex from tissues at 40–52 weeks gestation (A) and 33 weeks gestation (B) (glomerular generations formed and C) (glomerular generations formed). Scale bar = 100 μm. Linear regression analyses of the number of glomerular generations formed within the kidney versus gestational age (D), kidney weight (E), and body weight (F) in female (G) and male (H) fetuses and newborn infants. (From Ivan D, Scharfetter MS, Flores TL, et al. Development of the human fetal kidney from mid to late gestation in male and female infants. *atlasAnatomy*. 2016;27:275–283.⁷⁷)

Fig_20_3

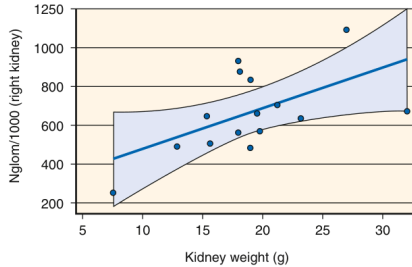


Fig. 20.6 Relationship between nephron number and mass of the right kidney in Caucasian infants aged ≤3 months who died within the first 3 months of life. $Ngrom/1000$, Number of glomeruli per kidney in 1000s. (From Zhang Z, Quinlan J, Hoy VV, et al. A common RET variant is associated with reduced newborn kidney size and function. *J Am Soc Nephrol*. 2008;19:2027–2034.⁹⁰)

Fig_20_6

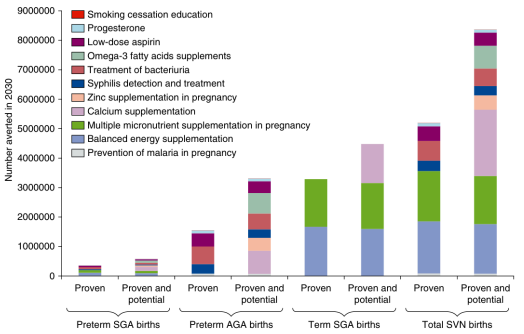


Fig. 20.9 Impact of maternal interventions on averting the risk of birth of small and vulnerable neonates from 81 low- and middle-income countries. AGA, Appropriate weight for gestational age; SGA, small-for-gestational age; SVN, small and vulnerable neonate. (Reprinted with permission from Elsevier, from Hofmeyr et al. Evidence-based antenatal interventions to reduce the incidence of small vulnerable newborns and their associated poor outcomes. *Lancet*. 2023; 401(10389):1733–1744.)

Fig_20_9

20. Early Life Origins of Hypertension and Chronic Kidney Disease · assets 10-18 / 33

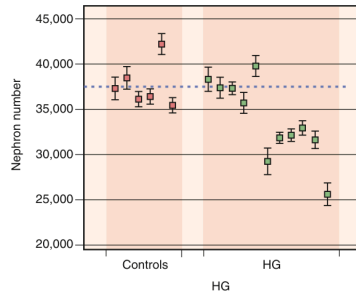


Fig. 20.10 Effects of maternal hyperglycemia on nephron number in rat offspring. *Dotted line* represents mean value in control group. *HG*, Hyperglycemia. (From Amiri K, Freund N, Vilar J, et al. Adverse effects of hyperglycemia on kidney development in rats: in vivo and in vitro studies. *Diabetes*. 1999;48:2240–2245.²³¹)

Fig_20_10

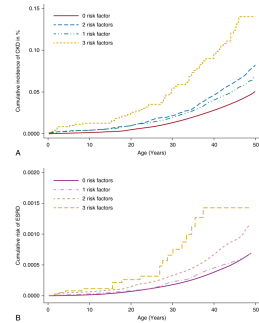


Fig 20.53 | (A) Cumulative incidence (%) of chronic kidney disease (CKD) according to number of birth-related risk factors (birth weight <10th percentile, birth weight for gestational age <10th percentile, and preterm birth before 37 weeks' gestation). (B) Cumulative incidence (%) of chronic kidney disease (CKD) according to number of birth-related risk factors (birth weight <10th percentile, birth weight for gestational age <10th percentile, and preterm birth before 37 weeks' gestation). Compared with having none of the exposures, individuals with one exposure had a significantly higher risk of CKD but not ESKF. Individuals with two exposures had a higher risk than individuals with one exposure, and individuals with three exposures had the highest risk for both CKD and ESKF. ESKF, End-stage renal disease; ESKF, From Glende A, Hasselzer JL, Shurus R, et al. Intrauterine growth restriction and risk of diverse forms of kidney disease during the first 50 years of life. *CJASN*. 2020;12:1413–1423 and Glende LLB, Mori AP, et al. Intrauterine growth restriction, preterm birth and risk of end-stage renal disease during the first 50 years of life. *Nephrol Dial Transplant*. 2020;35(7):1157–1167.

Fig_20_13

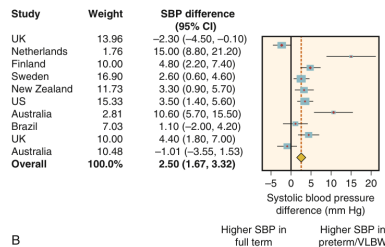


Fig. E2.2 Relationship among birth weight, preterm birth, and blood pressure. (A) Meta-analysis of odds for hypertension (*HTN*) in individuals with birth weights below 2500 g (low birth weight, *LBW*) compared with birth weights above 2500 g. The pooled odds ratio is shown as a diamond. (B) Meta-analysis of difference in systolic blood pressure (*SBP*) between individuals born preterm or very low birth weight (*VLBW*) compared with full term. Pooled SBP difference is indicated by the diamond and dashed vertical line. (A from Mu et al. *Arch Cardiovasc Dis*. 2012;105:99–113; see original paper for full references.⁴⁰⁸ B from de Jong F et al. *Hypertension*. 2012;59:226–234.⁴⁰⁹)

Fig_e20_2

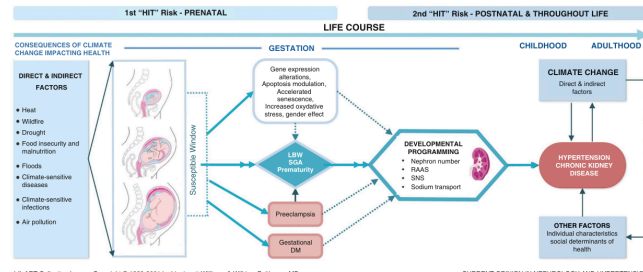


Fig. 20.11 Schematic illustrating the potential far-reaching impact of climate change on maternal health and kidney development during pregnancy and superimposed kidney injury (second hits) in later life. DM, Diabetes mellitus; LBW, low birth weight; RAAS, renin-angiotensin-aldosterone system; SGA, small for gestational age; SNS, sympathetic nervous system. (From Alvarez-Ellas AC, Brenner BM, Luyckx VA, et al. Climate change and its influence in nephron mass. *Curr Opin Nephrol Hypertens*. 2024;33[1]:102–109.)

Fig_20_11

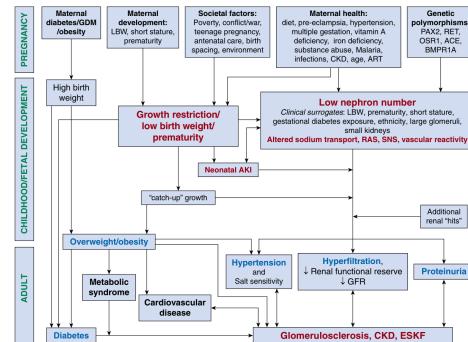
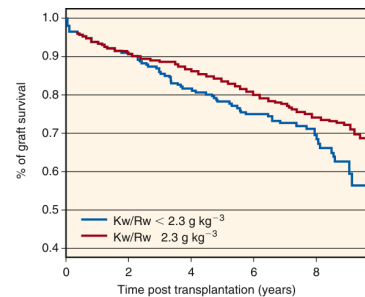


Fig. 20.14 Proposed mechanisms of fetal programming of hypertension and kidney disease. ART, Assisted reproductive therapies; CKD, chronic kidney disease; ESKD, end-stage kidney disease/failure; GDM, gestational diabetes mellitus; GFR, glomerular filtration rate; LBW, low birth weight; RAS, renin-angiotensin system; SNS, sympathetic nervous system. (Adapted from Luyckx VA, Bertram JF, Brenner BM, et al. Effect of fetal and child health on kidney development and long-term risk of hypertension and kidney disease. *Lancet*. 2013;382:273–283.¹⁴)

Fig_20_14



eFig. 20.3 Correlation between donor kidney weight (*Kw*)-to-recipient body weight (*Rw*) ratio and long-term graft survival in renal transplantation. Graft survival declines faster with a *Kw/Rw* <2.3 g/kg (*P* = .016). (From Giral et al. *J Am Soc Nephrol*. 2010;21:1022–1029 [690](#))

Fig_e20_3

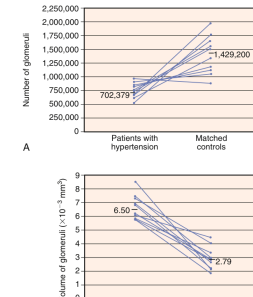
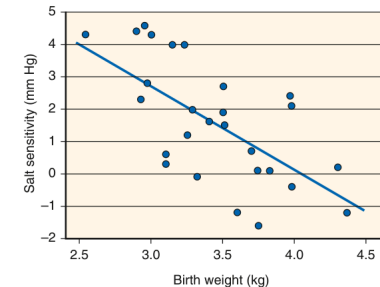


Fig. 20.12 (A) Nephron number, and (B) glomerular volume in Caucasian people with primary hypertension compared with normotensive controls. (From Keller G, Zimmer G, Mall G, et al. Nephron number in patients with primary hypertension. *N Engl J Med*. [2003] 348:101–108. [100](#))

Fig_20_12



eFig 20.1 Correlation between birth weight and salt sensitivity in 27 normotensive adults. Salt sensitivity of blood pressure was defined as the difference in mean arterial pressure on high-salt diet (200 mmol/day) compared with low-salt diet (60 mmol/day). $R = -0.06$; $P = .002$. (From de Boer et al. *Hypertension*. 2008;51(4):928-932.³⁶⁴)

Fig_e20_1

[illegible]

Table_20_1

20. Early Life Origins of Hypertension and Chronic Kidney Disease · assets 19-27 / 33

[illegible]

Table_20_2

Programmed Changes in Kidneys	Clinical Correlates for Programmed Risk
- Kidney size - Nephron number - Podocyte number - Glomerular volume - Maturation of glomeruli - Alterations in angiogenesis - Tubular structure - Tubular sodium transporter expression and/or activity - Altered renal vascular reactivity - Alterations in renin-angiotensin system - Alterations in sympathetic nervous system activity - Accelerated senescence, especially after catch-up growth - Predisposition to inflammation and fibrosis	- Low birth weight^{128,134} - Small for gestational age/intrauterine growth restriction^{131,132,136,138} - Premature birth^{89-91,134,142,143} - Neonatal acute kidney injury¹⁴⁴⁻¹⁵⁵ - Low kidney mass^{76,96} - Reduced kidney volume^{136,142,145} - Glomerulomegaly^{127,128,131,439} - Female sex⁹⁰ - Ethnicity: Aboriginal Australian^{349,448,450} - Older age^{7,57} - Shorter adult height^{77,840}

Table 20_5

Birth Weight Mean (Range), kg	N	Number of Glomeruli^a	Mean Glomerular Tuft Volume ($\mu\text{m}^3 \times 10^6$)	Total Glomerular Tuft Volume (cm^3)
2.65 (1.81-3.12)	29	770,860 (658,757-882,963)	9.2	6.7
3.27 (3.18-3.38)	28	965,729 (885,714-1,075,744)	7.2	6.8
3.93 (3.41-4.94)	30	1,005,356 (900,094-1,110,599)	6.9	6.6

^aAdjusted for age, gender, race, and body surface area.
 From Hoy WE, Hughson MD, Bertram JF, et al. *J Am Soc Nephrol*. 2005;16(9):2557-2264.

Table_20_3

	LBW	SGA	Preterm	HBW	Gestational DM	Preeclampsia	Maternal Overweight/Obesity	Maternal Vitamin A Deficiency	Rapid Catch-up Growth/Overweight	Low Nephron Number	Smaller Kidney Size/Weight	Increased Glomerular Volume
Increased blood pressure	✓	(✓)	✓	(✓)	✓	✓	✓		✓	✓	✓	✓
Salt sensitivity	✓				a					a	✓	
Reduced GFR	✓	✓	✓	✓	a				✓	a	✓	
Reduced renal functional reserve	no	✓	✓	✓								
Proteinuria	✓		✓	✓	✓				✓	a	✓	
Acute kidney injury (nephrites)	✓	✓	✓	✓		Protective				✓	✓	
Chronic kidney disease	✓	✓	✓	✓		✓	✓		✓	a	✓	✓
End-stage kidney failure	✓	✓	✓	✓	✓	✓					✓	✓
Transplant outcomes	✓										✓	✓
Death	✓								a		✓	
Increased glomerular volume	✓		✓		a				✓	✓		
Smaller kidney size/weight	✓	✓	✓				✓					
Congenital obstruction	✓			NS	✓		✓			a		
Renal dysplasia/aplasia	✓			NS	✓		NS			✓		✓

✓, Evidence from human studies; (✓), evidence suggested; a, evidence from animal studies only; DM, diabetes mellitus; LBW, low birth weight; NS, not significant; SGA, small-for-gestational age.

Table 20_6

	USA		Australia		Canada		Germany			
Population	Black Donors	White Donors	Indigenous Donors	Non-Indigenous Donors	Indigenous Donors	White Donors	BW ≤2.5kg	BW >2.5kg	Nephron number* 10% for age	Nephron number* 10% for age
Donor number	12,387	71,769	22	28	38	76	18	73	228	445
Putative LBW, programmed kidney	ref	LBW	ref	ref	HBW (offspring of diabetic pregnancies)	ref	18	BW	unknown	ref
Hypertension	–	–	50% ^a	6%	42%	19%	39%	15%	1	–
Proteinuria	–	–	81%	6%	21%	4%	81%	35%	1	–
IGFR	–	–	81%	38%	Not different	–	Not different	–	OR 2.72 (1.69–4.36)	ref
ESKF	74.4 per 10,000 vs. 23.9 per 10,000 in non-donors	22.7 per 10,000 vs. 5.9 per 10,000 in non-donors	19%	0%	1	0	0	0	0	0
Follow-up (years) (IGR 3.9–11.5)	7.6	7.6	16.1 (12.27–20.2)	6.37 (2.94–21.2)	14.6 ± 9.3	13.4 ± 9.5	≥5	≥1–3	10.5	–
Reference	782	782	663	666	666	666	668	668	77	–

BW, Birth weight; ESKF, end-stage kidney failure; IGFR, glomerular filtration rate; HBW, high birth weight; LBW, low birth weight; OR, odds ratio; ref, reference group.

Table_20_9

[illegible]

Table_20_4

Table 2a. Maternal Factors Associated With Increased Risk of Small and Vagile Neon Births			
	Maternal Factor	Global Incidence/Prevalence (%)	Specific Incidence/Prevalence
Developmental	Maternal weight < 42.5 kg or > 45 kg	19.7-18.8%	626/ 671
	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671
	Maternal weight < 150 lb or > 150 lb	19.7-18.8%	626/ 671
	Maternal weight < 68 kg or > 68 kg	19.7-18.8%	626/ 671
Behavioral	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671
	Maternal weight < 150 lb or > 150 lb	19.7-18.8%	626/ 671
	Maternal weight < 68 kg or > 68 kg	19.7-18.8%	626/ 671
	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671
Demographic	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671
	Maternal weight < 150 lb or > 150 lb	19.7-18.8%	626/ 671
	Maternal weight < 68 kg or > 68 kg	19.7-18.8%	626/ 671
	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671
Health-related	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671
	Maternal weight < 150 lb or > 150 lb	19.7-18.8%	626/ 671
	Maternal weight < 68 kg or > 68 kg	19.7-18.8%	626/ 671
	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671
Social	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671
	Maternal weight < 150 lb or > 150 lb	19.7-18.8%	626/ 671
	Maternal weight < 68 kg or > 68 kg	19.7-18.8%	626/ 671
	Maternal weight < 100 lb or > 100 lb	19.7-18.8%	626/ 671

Table 20_8

[illegible]

Table_e20_1_part1

[illegible]

Table_e20_1_part2

[illegible]

Table_e20_1_part3

Clinical Circumstance	Population/Age	Glomerular Number/Kidney Volume (% change)	Birth Weight	Blood Pressure	Kidney Function/Pathology
Hypertensive vs. normotensive Japanese ⁴⁴	Adult males	1.39 (nonsteroidic + (sclerotic) ~ kidney volume	NA	NA	1 SNGFR 1 eGFR with 1 nephrons
Hypertensive vs. normotensive Aboriginal Australian ⁴⁹	Adult	NA	NA	NA	Glomerular hypertrophy
Aboriginal Australians vs. Caucasian Australian ⁵⁰	Adult	1.30	1	NA	Glomerular hypertrophy
Senegalese Africans ^{46,52}	5-70 years	NA	NA	NA	1 Variability of glomerular size with 1 glomerular numbers
Genetic polymorphisms:					
RET1(476A) polymorphism ⁵¹	Newborns	1.10	NA	NA	NA
PAX2 AAA haplotype ⁵⁴	Newborns	1.10	NA	NA	NA
Combined RET1(476A) polymorphism and PAX2 AAA haplotype ⁵¹	Newborns	1.23	NA	NA	NA
I/D ACE polymorphism ⁵⁶	Newborns	1.8	NA	NA	NA
BAHPI1 ^{70222G} polymorphism ⁵⁵	Newborns	1.13	NA	NA	NA
OSR1(123590G) polymorphism ⁵⁷	Newborns	1.12	NA	NA	NA
Combined OSR1 and RET polymorphisms ⁵⁷	Newborns	1.22	NA	NA	NA
Combined OSR1 and PAX2 polymorphisms ⁵⁷	Newborns	1.27	NA	NA	NA
ALDH1A2 ^{78039G} polymorphism ⁵³	Newborns	1.22	NA	NA	NA

Table_e20_1_part4

Maternal Risk Factor for Preterm Birth	Animal		Human	
	BP	Nephron Number/Kidney Size	BP	Nephron Number/Kidney Size ^a
Vitamin D deficiency	↑	↑ (↑ immature glomeruli)	ND	ND
Anemia/iron deficiency	↑	↑	ND	ND
Smoking	ND	↓	↑	↓
Alcohol intake	ND	↓	ND	ND
Antibiotic use (UTI)	ND	↓	ND	ND
Chorioamnionitis	ND	↓	ND	ND
Steroids	↑	↓	Normal	Not different
Cyclosporine/tacrolimus (maternal transplant)	↑	↓	ND	Unknown
Maternal dialysis	ND	ND	ND	Some ↓
Maternal diabetes	↑	↓	↑	ND
Preeclampsia	ND	↓	↑	ND

^aKidney size generally determined by ultrasound.

BP, Blood pressure; ND, not determined; UTI, urinary tract infection; ↑, increased; ↓, decreased.

Compiled from references: ^{214,225,227,235,237,240,242,243,245,246,247}

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[illegible]

Table_e20_3_part1

Experimental Model	Possible Mechanism of Nephron Number Reduction	References
Ureteral obstruction: postnatal	- Cell proliferation - Apoptosis of tubular cells - Delayed maturation of interstitial fibroblasts → interstitial fibrosis	121
Preterm birth (and associated postnatal factors)	? Alteration of postinductive processes - Accelerated maturation of glomeruli - Early differentiation of progenitor cells - Altered HIF signaling - Impaired angiogenesis - Oxidative stress ? Factors associated with shift from intrauterine to extrauterine environment - Kidney injury and repair - Reduced telomere length	49, 167, 253, 278, 282, 626, 716

ERK, Extracellular signal-regulated kinase; *HIF*, hypoxia-inducible factor; *11-βHSD2*, 11-β-hydroxysteroid dehydrogenase; *IGF-2*, insulin-like growth factor 2; *MAPK*, mitogen-activated protein kinase; *NF-κB*, nuclear factor kappa B; *RAAS*, renin-angiotensin-aldosterone system; *VEGF*, vascular endothelial growth factor.

Table_e20_3_part2

Model	Specimen	Timing of blood sampling	Age and Sex of Subjects or Offspring of Parents	Offspring or Parents Genotype	Physiological Response	Reference(s)
Observational	Obese	Early postnatal	Obese D	Plasma levels of insulin, triglycerides, and HDL-C	↑ Fasting IGF-1 ↑ Plasma IGF-1	175
	Pub	8-7 years (2)	Pub	↓ Plasma IGF-1 ↓ HDL-C ↑ HDL-C	↑ Fasting IGF-1 ↑ HDL-C	183
	Pub	Mid to late gestation	Free	↑ Plasma IGF and IGF-1 ↑ HDL-C	NO	184
	Pub	Mid to late gestation	4 and 6 mo (2)	↑ Plasma IGF-1 ↑ HDL-C ↑ HDL-C ↑ HDL-C ↑ HDL-C	↑ Fasting IGF-1 ↑ HDL-C ↑ HDL-C ↑ HDL-C ↑ HDL-C	182

Table_e20_4